

Kaggle CMI Sensor Data: Transformation Assessment

Can the CMI Detect Behavior with Sensor Data be transformed for MediaPipe model training?
Prepared by: Nishanth Tiglao • ENEB453 / ENEB456 Combined Final Project • April 2026

VERDICT: NOT TRANSFORMABLE

After thorough technical analysis, the Kaggle CMI wearable sensor data **cannot be meaningfully transformed** into training data for a MediaPipe-based computer vision model. The sensor modalities are fundamentally incompatible. This is not a limitation of engineering effort — it is a **physics-level mismatch** between what each system measures.

1. What the Kaggle Data Contains

The CMI competition dataset ([Kaggle link](#)) collects data from a **wrist-worn wearable device** with multiple physical sensors:

Sensor	Columns	What It Measures	Physical Units
Accelerometer (3-axis)	acc_x, acc_y, acc_z	Linear acceleration of the wrist	g-force (m/s ²)
Rotation (Quaternion)	rot_w, rot_x, rot_y, rot_z	3D orientation of the wrist/forearm	Unit quaternion
Thermal (5 sensors)	thm_1 — thm_5	Skin/proximity temperature	°C (27-34 typical)
Time-of-Flight (5×64)	tof_1_v0 — tof_5_v63	Distance from wrist to nearby surfaces	Millimeters

Total: 341 columns across 574,945 rows, 81 subjects, 18 gesture classes.

2. What Our MediaPipe System Uses

Our BFRB detection system uses a webcam with **MediaPipe Holistic** to extract:

Landmark Set	Count	What It Measures	Units
Hand landmarks	21 points	2D positions of hand joints (wrist, knuckles, fingertips)	Normalized image coordinates (0–1)
Face mesh landmarks	468 points	2D positions of facial features (eyes, nose, lips, cheeks)	Normalized image coordinates (0–1)
Pose landmarks	33 points	Body skeleton positions	Normalized image coordinates (0–1)

3. Why Transformation Is Impossible: Point-by-Point Analysis

For Kaggle data to train our MediaPipe model, we would need to convert each sensor reading into an equivalent MediaPipe landmark feature. Here is why each mapping fails:

BLOCKER 1: Thermal Sensors → No Equivalent

The Kaggle device has 5 thermal sensors measuring **actual skin temperature** at the contact point. When a hand touches the face, thermal readings rise from ~28°C to ~33°C. This is the #1 most important feature in the Kaggle model (highest Gini importance).

MediaPipe cannot measure temperature. A webcam captures light, not heat. There is no proxy, no approximation, and no workaround. This signal simply does not exist in our system.

Result: Complete loss of the dataset's most discriminative feature.

BLOCKER 2: Time-of-Flight (320 columns) → Lossy Approximation

The ToF sensors emit infrared light and measure the **physical distance in millimeters** from the wrist to nearby surfaces (face, desk, etc.). They provide a 5×64 spatial grid — essentially a depth map from the wrist's perspective.

MediaPipe gives us hand-to-face **pixel distances in normalized image coordinates (0–1)**. These are:

- Affected by camera distance (person closer to camera = landmarks farther apart in pixels)

- Non-linear relationship to physical mm (varies with camera focal length, lens distortion)
- Missing the 3rd dimension (depth is unreliable in MediaPipe's z-axis estimation)
- Only 7 distance pairs vs. 320 ToF values — 97.8% information loss

Result: No valid mapping from mm-scale depth grids to 2D pixel ratios.

BLOCKER 3: Accelerometer → Approximate but Incompatible Scale

We *can* compute wrist velocity from MediaPipe by tracking `hand[0]` frame-to-frame. However:

- Kaggle measures **actual acceleration in g-force** (inertial measurement)
- MediaPipe gives **pixel displacement per frame** (visual measurement)
- A model trained on "acc_y = -9.8 means arm hanging" cannot use "wrist.y = 0.7 means wrist is at 70% of frame height"
- The relationship between pixel velocity and physical acceleration is non-linear and depends on camera setup

Result: Conceptually similar but numerically incompatible. Decision boundaries don't transfer.

BLOCKER 4: Quaternion Rotation → Different Reference Frame

Quaternions represent 3D wrist orientation relative to gravity. We *can* estimate palm orientation from MediaPipe hand landmarks, but:

- Kaggle quaternions are measured by a gyroscope — precise 3D orientation
- MediaPipe palm direction is estimated from 2D projected landmark positions — noisy, ambiguous
- The reference frames are completely different (IMU world frame vs. camera image frame)

Result: Qualitatively similar concept, quantitatively incompatible.

4. We Already Tried: v1 Results

In our v1 approach, we *did* attempt this transformation. We engineered 16 features to bridge the domains:

Kaggle Sensor → Engineered Feature → MediaPipe Proxy

Accelerometer (3-axis)	→ acc_mag_mean/std/max	→ Wrist velocity from hand[0] Δ
Quaternion (4-axis)	→ rot_speed_mean	→ Palm direction vector
Thermal (5 sensors)	→ thm_mean, thm_max	→ Hand-face overlap fraction ???
ToF grid 1 (64 values)	→ tof1_min_mean, tof1_mean	→ Wrist→Nose distance
ToF grid 2 (64 values)	→ tof2_min_mean, tof2_mean	→ Index→Mouth distance
ToF grid 3 (64 values)	→ tof3_min_mean, tof3_mean	→ Thumb→Cheek distance
ToF grid 4 (64 values)	→ tof4_min_mean, tof4_mean	→ Palm→Forehead distance
ToF grid 5 (64 values)	→ tof5_min_mean, tof5_mean	→ Wrist→Chin distance

Training Results (on Kaggle data only):

- Accuracy: 77% / AUC-ROC: 0.837 — decent on the source domain
- Top features by importance: thermal readings (27%), ToF grids (45%), accelerometer (20%)

Why it fails on real webcam data:

1. The model learned that thermal rise = BFRB contact. Webcam has no thermal signal.
2. The model learned ToF patterns in millimeters. MediaPipe distances are in 0-1 pixel ratios.
3. MinMax normalization does NOT fix this — it normalizes within each domain but the *decision boundaries* (what separates BFRB from non-BFRB) are defined in the source domain's scale.
4. This is a **zero-shot cross-domain transfer**: the model never saw a single MediaPipe feature during training. Without target-domain labeled data, transfer is impossible.

5. Feature-by-Feature Compatibility Matrix

Kaggle Feature	Importance	MediaPipe Equivalent?	Transferable?
Thermal (thm_mean, thm_max)	27%	None — cameras cannot measure temperature	NO
ToF 1-5 (10 features)	45%	Pixel distances (different scale, reference, and resolution)	NO
Accelerometer (3 features)	20%	Pixel velocity (conceptually similar, numerically incompatible)	PARTIAL
Rotation (1 feature)	8%	Palm direction (different reference frame)	PARTIAL

72% of model importance relies on features with no valid MediaPipe equivalent.

6. Why Even "Creative" Transformations Fail

6.1 Can we map ToF mm to pixel ratios with a scaling function?

No. The relationship between physical distance (mm) and pixel distance depends on: camera focal length, sensor size, user distance from camera, and lens distortion. These vary per user and per session. There is no single scaling function.

6.2 Can we use the labels and ignore the features?

The Kaggle dataset has labeled BFRB gestures (hair pulling, skin picking, etc.). We *can* use this knowledge to design rules — which is exactly what our v2/v3 rule-based engine does. But using labels without features is not "transforming the data for model training" — it's just reading the competition description.

6.3 Can we learn temporal patterns from the data?

BFRB temporal patterns (duration, frequency) are already encoded in our rule-based dwell timers. The Kaggle data would add no new information beyond what clinical literature provides.

6.4 Can we use domain adaptation techniques?

Domain adaptation (e.g., DANN, CycleGAN) requires *paired samples* from both domains or at least features in the same dimensionality and modality. Wearable sensor data and camera landmark data have no structural correspondence to align.

7. What We Did Instead: v3 Rule-Based Engine

Rather than forcing an impossible cross-domain transfer, we built a detection engine that works **natively on MediaPipe landmarks**:

- 4-state machine (IDLE → PROXIMITY → DWELL → BFRB) using face-width-normalized distances
- Pinch-pull engine for hair pulling and skin picking (2-finger pinch + pull-away)
- **v3 additions**: mouth aperture (eating filter), extended finger count, micro-motion jitter, palm orientation, dual-signal confidence scoring, eating/resting negative detectors

This approach avoids the domain gap entirely and delivers **real-time detection at 30fps** with significantly fewer false positives.

Conclusion

The Kaggle CMI dataset is valuable for understanding BFRB behaviors and validating that wearable-based detection works. However, the data was collected from **physical sensors (accelerometer, gyroscope, thermal, time-of-flight)** that measure fundamentally different quantities than what a **webcam + MediaPipe** system can observe.

The three critical blockers are:

1. **Thermal data has no camera equivalent** (72% of the model's top features depend on sensors that don't exist in our system)
2. **Physical units (mm, g-force, °C) cannot be mapped to normalized pixel coordinates** without camera calibration data that varies per user
3. **Zero-shot cross-domain transfer requires structural feature alignment** that does not exist between these two modalities

The correct engineering decision — which we have implemented — is to use MediaPipe's native landmark data directly with rule-based heuristics, leveraging the *behavioral knowledge* from BFRB research (including insights from the Kaggle competition description) without attempting to force incompatible sensor data through a model.